

Andrija Mitrović

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EDUCATION

B.Sc. Computer Science

2018 – Present

University of Kragujevac, Faculty of Natural Sciences and Mathematics

Relevant coursework: Algorithms & Data Structures, Operating Systems, Databases, Software Engineering, Computer Networks

EXPERIENCE

Back End Engineer Intern

July 2024 – August 2024

ASEE

- Developed a REST API for a Personal Finance Management application using **.NET** and **PostgreSQL**, enabling users to create, log, and categorise financial transactions by category.
- Implemented filtering endpoints for transaction history and financial reporting, helping users review spending patterns and retrieve relevant records more efficiently.
- Containerised the application with **Docker** and validated all API endpoints using **Postman**, ensuring the service worked reliably and could be run consistently across environments.

Full Stack Engineer Intern

July 2023 – August 2023

Inovatec

- Collaborated within a 4-person team to develop an internal office asset management tool, streamlining hardware inventory tracking across multiple offices.
- Enabled staff to efficiently report broken or missing items, ensuring an accurate and up-to-date record of office assets.
- Engineered the backend using **Spring Boot**, utilized **PostgreSQL** for data management, and implemented the frontend with **React**.

PROJECTS

FitTrack (2026) | fitcadence.net

- Built a full-stack workout logging application allowing users to record, track, and review training sessions, featuring JWT-based authentication and authorization.
- Developed a **Java Spring Boot** backend with a **PostgreSQL** database and a **React** frontend, with unit and integration testing using **JUnit** and **Mockito**.
- Implemented end-to-end testing with **Selenium** to validate key user flows such as authentication, workout creation, and session review.
- Deployed to fitcadence.net using Dockerized services, a **Spring Boot** backend server and a **React** app served via **Nginx**, secured with HTTPS.
- Established a CI/CD pipeline with **GitHub Actions** to automatically run tests and deploy the application on every push.

Grep Lite (2026) | andrijamitrovic.github.io/haskell-grep-lite

- Developed a lightweight grep-like text search tool in **Haskell** to find lines matching user-provided regular expressions.
- Implemented a custom regex parser converting patterns into an abstract syntax tree (AST), supporting key regex features such as concatenation, alternation, repetition, and grouping.
- Applied Glushkov NFA construction to transform the regex AST into an epsilon-free nondeterministic finite automaton (NFA) for efficient pattern matching.
- Extended the project for browser use by compiling the Haskell regex engine to **WebAssembly**, enabling client-side regex matching without a backend server.

SKILLS

Languages: C#, Java, TypeScript, SQL

Frameworks: Spring Boot, .NET, React, Angular

Databases: PostgreSQL

Tools & DevOps: Docker, Git, GitHub Actions, Postman, REST APIs, JWT

Certifications: C1 Advanced English - Cambridge Final Exam, British Council Belgrade